

Jeet Vora

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Research Interests

Computer Vision, Video Understanding, Multi-View Geometry, Spatio-Temporal Learning, 3D Vision, Dynamic 4D Scene Understanding, World Models, Generalization

Education

Master's By Research in Computer Science *IIIT Hyderabad* **CGPA: 8.2/10.0** 2019-2023
(Specialization in Artificial Intelligence and Robotics)
Bachelor of Engineering in Computer Science *University of Mumbai* **CGPA: 8.96/10.0** 2015-2018
Diploma in Computer Science *Shri.Baghubhai Mafatlal Polytechnic* **CGPA: 8.2/10.0** 2012-2015

Publications

- Jeet Vora**, Dutta, S., Jain, K., Karthik, S. & Gandhi, V. Bringing Generalization to Deep Multi-View Pedestrian Detection in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops* [Paper Link](#) (Jan. 2023), 110–119.

Research and Work Experience

Research Consultant *King Abdullah University of Science and Technology (KAUST)* Nov 2025 – Present

- Research Collaboration:** IVUL Lab, KAUST - Prof. Bernard Ghanem's group; primarily collaborating with Dr. Silvio Giancola, Dr. Karen Sanchez, Dr. Meryem Ramazanova, and Jintao Ma.
- Project Page:** <https://github.com/OpenSportsLab/opensportslib>
Documentation: <https://opensportslab.github.io/opensportslib>
PyPI: <https://pypi.org/project/opensportslib>
- Leading the research, architecture, and development of OpenSportsLib, a large scale sports video understanding framework focused on spatio-temporal AI modeling and multimodal perception.
- Architecting a modular and extensible Python research library supporting Action Classification, Temporal Localization, Retrieval, and Dense Video Description.
- Developing scalable training, inference, and evaluation pipelines for multi-modal inputs, including raw video, frame sequences, pre-extracted features, and player tracking data.
- Establishing reproducible experimentation protocols and benchmarking standards for standardized evaluation across large annotated sports datasets.
- Conducting empirical studies on model generalization, efficiency, and performance trade-offs to advance research in sports video analytics.

Research Engineer *Animaker India Pvt. Ltd.* July, 2022 - Nov, 2025

- Led applied research in AI-ML (CV/NLP) across multiple Animaker platforms (Animaker, Steve.ai, Picmaker, Vmaker), delivering AI systems serving 25M+ users.
- Designed and optimized deep learning models for video/image matting, multimodal retrieval, script-to-Animated video generation, script-to-GenAI video architecture, controllable character builder pipeline, and audio driven talking-head animation.
- Implemented knowledge distillation and model compression techniques achieving 8× inference speedup, enabling 4 concurrent model workers on a single 24GB GPU.
- Architected end-to-end ML pipelines including dataset curation, training infrastructure, evaluation protocols, CI/CD integration, and GPU optimized scalable deployment.
- Drove cross-functional collaboration with Product, DevOps, and Engineering teams to transition research prototypes into reliable, production grade AI services.

Research Student *IIIT Hyderabad* Jan, 2020 - May, 2023

- Advisor:** CVIT Lab, IIITH - Dr. Vineet Gandhi;
- Thesis:** [Towards Generalization in Deep Multi-View Pedestrian Detection and Tracking](#).
- Proposed a multi-view pedestrian detection framework explicitly designed for cross-camera generalization across unseen scenes and varying camera configurations.
- Achieved a 20% improvement in MODA and MODP metrics over benchmarks.
- Created synthetic datasets for multi-camera detection using GTA-V and Unity game engines.
- Developed post-processing tools for joint calibration and synchronization of multiple cameras.
- Demonstrated strong generalization from synthetic to real-world scenarios (Sim2Real).

Research Student *Star Sports*

Jan, 2020 - May, 2023

- **Project Supervisor:** CVIT Lab, IIITH - Dr. Vineet Gandhi;
- **Blog:** [IIITH's Player Tracking Tech Drives Asia Cup's Narrative For Viewers.](#)
- Developed **Real-Time Multi-Camera Player Tracking** system operating at 30fps on 4K/FullHD live broadcast streams.
- Modeled multi-view player identity fusion as an optimization problem using linear assignment in a unified bird's-eye representation.
- Developed a robust multi-camera calibration and homography estimation framework for large scale cricket stadiums, enabling geometrically consistent bird's-eye view projection and real-world trajectory analysis.
- Engineered end-to-end deployment infrastructure using Blackmagic Design hardware for 4K live streaming and GPU based processing.
- Successfully deployed and broadcasted during Asia Cup 2022 (UAE), Asia Cup 2023 (Sri Lanka), and Tamil Nadu Premier League (TNPL).

Research Collaboration *Apple*

Jan, 2022 - Mar, 2022

- **Project Supervisor:** CVIT Lab, IIITH - Dr. Vineet Gandhi.
- Developed a video understanding prototype for industrial assembly-line monitoring using deep learning based **Video Classification and Action Recognition with Temporal modeling techniques**.
- Designed and evaluated spatio-temporal models to classify fine-grained assembly operations under real-world production constraints.

Software Engineer *Vistaar Technologies, Inc.*

Jun, 2018 - Aug, 2019

- Worked as part of the Reimbursement Solution Team, analyzing and providing technical solutions.
- Directly involved in development, delivery and onboarding of Vistaar's SaaS based Reimbursement product for clients.
- Designed and implemented tailored solutions for Diageo, E. & J. Gallo Winery, and Brown-Forman, improving and streamlining processes, maximizing recovery, and enhancing customer success.

Teaching and Mentorship

AI-ML Mentor *InLustro Learning*

July, 2024 – Sept, 2025

- Providing industry-relevant AI-ML training and mentorship to students, professionals, and faculty members through lecture sessions and hands-on workshops.
- Mentored 14 AI-ML projects, guiding teams in model development, optimization, and deployment.
- Collaborated with clients, including L&T Madh Training Academy and MLR Institute of Technology, to deliver tailored AI-ML solutions.

Teaching Assistant *IIIT Hyderabad*

Jan, 2022 - May, 2022

- **Lecturer:** Dr. Vineet Gandhi, **Course:** Statistical Methods in AI (SMAI), Spring 2022.
- Prepared assignments, conducted lab sessions/tutorials, and guided students through projects.

AI-ML Mentor *TalentSprint*

Oct, 2021 – Oct, 2022

- **Supervisor:** Dr. Anoop Namboodiri.
- Mentored AI-ML projects as part of the AI/ML program by IIIT Hyderabad and TalentSprint.
- Provided guidance on AI-ML projects and conducted lab sessions.
- Supervised 4 teams working on Image Tagging and Road Object Detection.

Technical Skills

Programming: Python, C++, C, SQL

Core Expertise: Computer Vision, Multi-Object Tracking, Multi-View Geometry, Video Understanding, Spatio-Temporal Modeling, Generative Models, Domain Generalization

Deep Learning Frameworks: PyTorch, ONNX, TensorRT

Model Optimization & Acceleration: Knowledge Distillation, Quantization, AWS Inferentia

Vision & 3D Libraries: OpenCV, Open3D

Deployment & Systems: Docker, TorchServe, AWS SageMaker, AWS EC2, Flask

Data & Experimentation: Pandas, Elasticsearch, Kibana

Game Engines / Simulation: Unity, GTA-V (Synthetic Data Generation)

Tools: Git, LaTeX

Courses

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| • Statistical Methods in AI | • Computer Vision | • Optimization Methods |
| • Digital Image Processing | • Mobile Robotics | • Topics in Applied Optimization |